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# Lesson 3: Regression

Technologies for Artificial Intelligence

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# Before we start

- Exercise 2, which we discuss now
- The pipeline you built is the one we fit models into from now on

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# Today: the first real model

- The model and what it claims
- The exact solution, and why it sometimes fails
- Gradient descent
- Curves, and the price of flexibility
- Ridge and Lasso

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# One dataset, all lesson

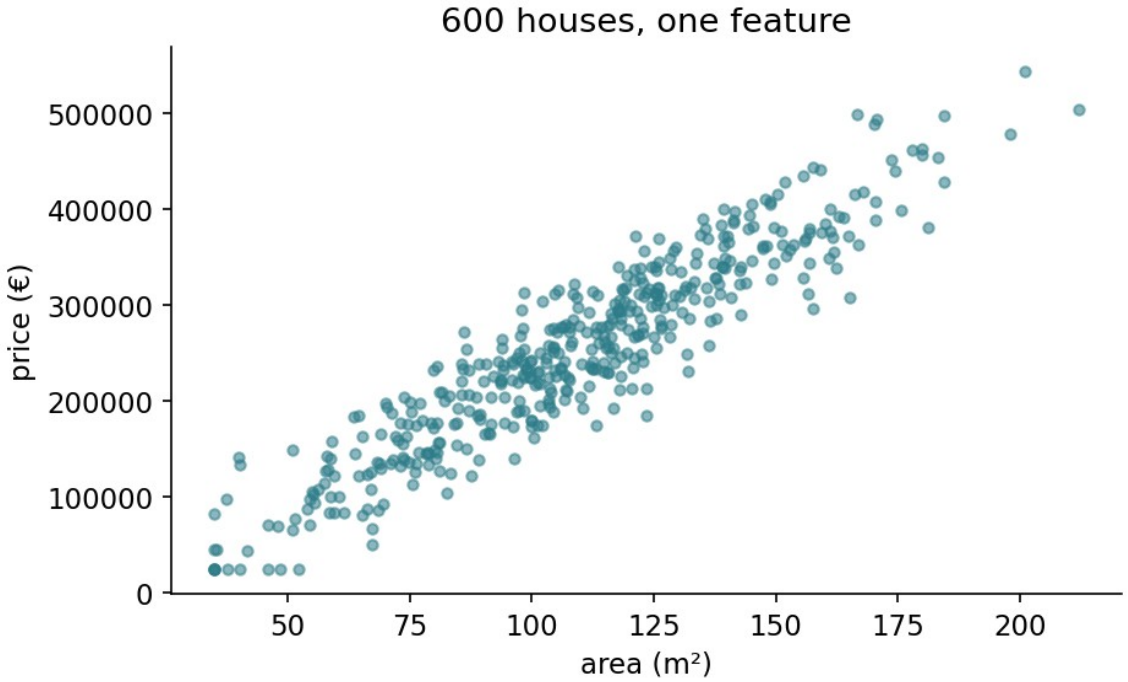
600 houses. Six measurements. A price.

**And we know the coefficients that generated it.**

In code: `train_test_split(X, y, test_size=0.25)` keeps 150 houses for testing.

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# What we are trying to predict




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# What a linear model claims

Each feature contributes a fixed amount per unit, and the contributions add up: $\hat{y} = w_1x_1 + w_2x_2 + \ldots + w_nx_n + b$, one coefficient w per feature plus an intercept b.

- Every square metre: the same 2,400 €
- Every kilometre out: the same −6,500 €

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# Fitting means minimising something

We need one number saying how badly a candidate model is doing.

Adding up the errors, $\Sigma_i (\hat{y}^{(i)} - y^{(i)})$, fails: +50,000 and −50,000 cancel to zero.

Every error must count as a positive amount:

- absolute error $\Sigma_i |\hat{y}^{(i)} - y^{(i)}|$
- squared error $\Sigma_i (\hat{y}^{(i)} - y^{(i)})^2$

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# Why squared, not absolute

- Squared error $\Sigma_i (\hat{y}^{(i)} - y^{(i)})^2$: each error $e = \hat{y}^{(i)} - y^{(i)}$ counts as $e^2$, not as $|e|$
- **Differentiable everywhere**: $e^2$ is smooth, $|e|$ has a corner at $e = 0$
- **Punishes one large error more**: double an error and $|e|$ doubles, $e^2$ quadruples
- Under Gaussian noise it is **maximum likelihood**: it picks the coefficients that make the observed prices most probable

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# The cost function: mean squared error

$$J(w,b) = \frac{1}{2m} \sum_{i=1}^{m} \left(\hat{y}^{(i)} - y^{(i)}\right)^2$$

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# Trying $\hat{y} = 2{,}400 \cdot \text{area} + 45{,}000$

| Area ($m^2$) | True (€) | Predicted (€) | Error $\hat{y} - y$ | Error$^2$ |
|---|---|---|---|---|
| 80 | 240,000 | 237,000 | −3,000 | $9.0 \times 10^6$ |
| 120 | 320,000 | 333,000 | +13,000 | $1.69 \times 10^8$ |
| 200 | 540,000 | 525,000 | −15,000 | $2.25 \times 10^8$ |

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# Squared error spends its attention on the worst house

- The 3,000 € error contributes about **2%** of the total
- The two large errors contribute the other **98%**
- Halving the 15,000 € error cuts the cost by **42%**
- In code: `mean_squared_error(y, y_pred)`, the mean of the squared errors without the ½

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# A cost needs something to be compared against

- Trivial answer: **always predict the mean**, `DummyRegressor()`
- Root mean squared error, RMSE = √MSE, on the 150 test houses: **96,440 €** for the mean, **20,341 €** for the model
- $R^2 = 1 - \Sigma(y - \hat{y})^2 \div \Sigma(y - \bar{y})^2 =$ **0.956** (`r2_score`): 0 is the mean, 1 is perfect
- Quote the euros too: $R^2$ depends on *this* test set's variance

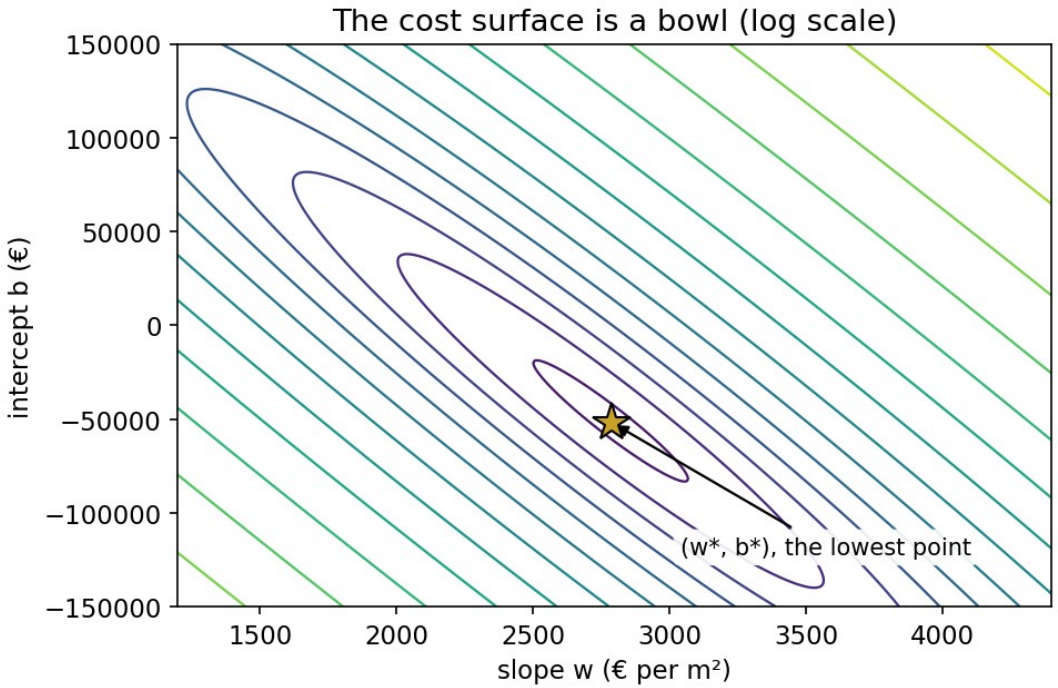
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# Fitting is finding where J is smallest

$$(w^*, b^*) = \arg\min_{w,\ b} J(w, b)$$

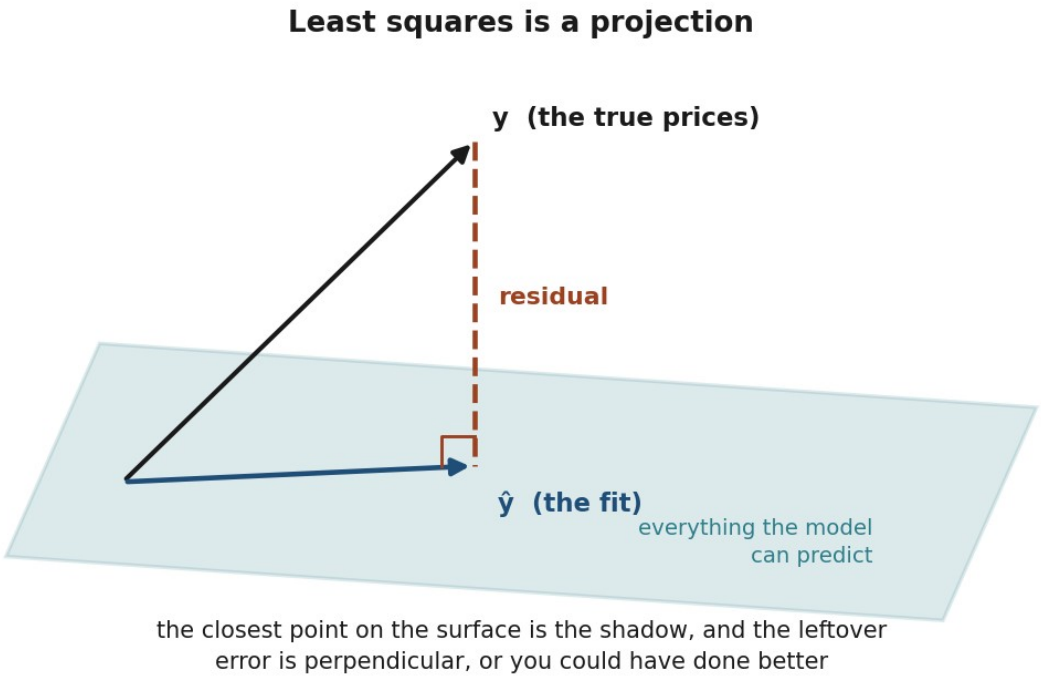
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# The cost is a bowl




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# The picture behind the exact solution




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# θ and X: all parameters, all houses

- $\theta = (b, w_1, \ldots, w_n)^T$: every parameter in one column, the intercept first
- X: one row per house; a column of 1s, then one column per feature
- So $X\theta$ gives $b + w_1x_1 + \ldots + w_nx_n$ for every house at once: all m predictions in one product
- The cost becomes $J(\theta)$, and the next slide solves for θ

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# The normal equation

$$\theta = (X^\top X)^{-1} X^\top y$$

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# The same equation, by hand

| Quantity | Value |
|---|---|
| the three houses | 80, 120, 200 m² → 240k, 320k, 540k |
| X, and θ | a column of 1s (for b) beside the areas; θ = (b, w) |
| $X^TX$ | rows (3, 400) and (400, 60,800) |
| $X^Ty$ | (1,100, 165,600) |
| determinant | 22,400 |
| **slope w** | **2,536 €/m²** (truth 2,400) |
| intercept b | 28,600 € |

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# And it really is the minimum

- The second derivative is $X^TX$, never negative in any direction
- So the cost is **convex**: one bottom, no local traps
- Any stationary point (zero gradient) is *the* answer, not *an* answer
- In code: `LinearRegression().fit(X, y)` solves the same problem

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# So why learn anything else?

- $(X^TX)^{-1}$ costs about $n^3$ operations
- Two columns carrying one fact → **no inverse exists**
- Nearly-redundant columns → an inverse that is useless

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# How stretched is the valley?

The **condition number** κ = steepest curvature ÷ shallowest curvature: how many times steeper the steepest direction of the bowl is than the shallowest.

- Housing features as recorded: **285**
- The same features, standardised: **3.4**
- Add `area_sqft` beside `area_sqm`: **2,286**

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# The picture behind gradient descent

You are on a hillside in fog.

You cannot see the valley, but you can feel which way the ground slopes.

So: step against the slope, by an amount set by the **learning rate** $\alpha$, and repeat until the cost stops falling.

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# The update rule

$$w \leftarrow w - \alpha \frac{\partial J}{\partial w}, \qquad \frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_i \left( \hat{y}^{(i)} - y^{(i)} \right) x_j^{(i)}$$

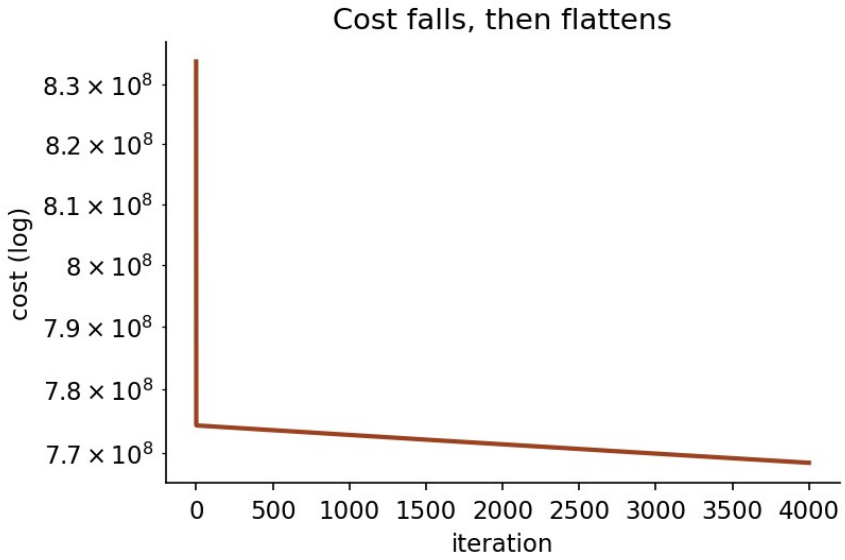
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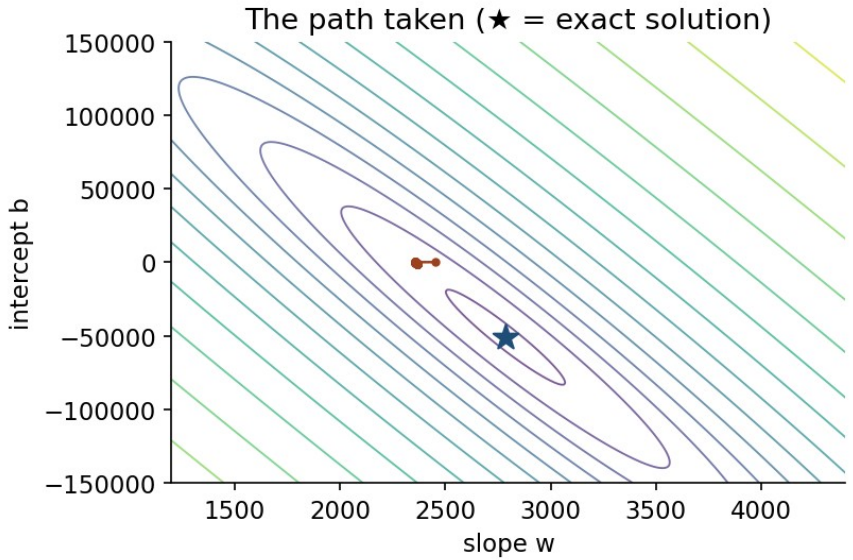
# One step on two houses, from w = 0 and b = 0

| 80 $m^2$ at 240k, 200 $m^2$ at 540k; $\alpha = 10^{-5}$ | Gradient | After one step |
|---|---|---|
| slope w | −63,600 | 0.636 |
| intercept b | −390 | 0.0039 |

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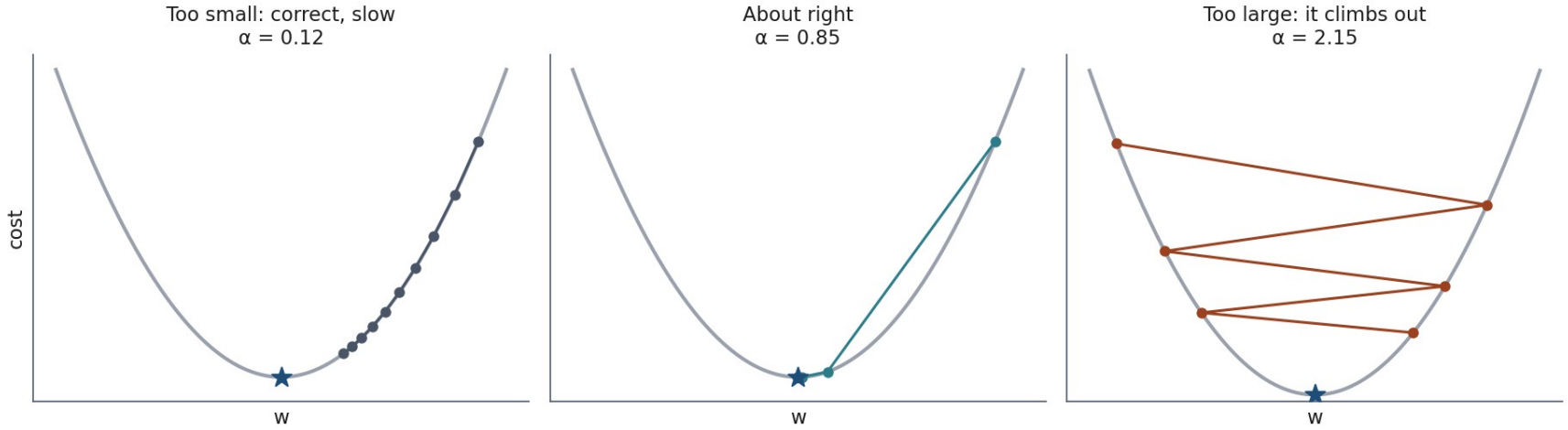
# The path it takes




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# Three learning rates




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# Notebook 1, live

| section, and its slides | you call (→ the number to come back with) |
|---|---|
| one feature, 4-5 | `train_test_split` → 450 houses to fit, 150 to test |
| the exact answer, 15-19 | the normal equation in NumPy, then `LinearRegression` → 2,785 €/m², agreeing to 1.82e-12 |
| the iterative answer, 22-26 | NumPy only: gradient descent → the same answer, slowly |
| all six features, 51-52 | `LinearRegression` → area 2,421, bedrooms −16.6% |
| the baseline, 12 | `DummyRegressor(strategy="mean")`, `r2_score` → 96,440 € against 20,341 €, $R^2$ 0.956 |

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# Break

# What if the relationship bends?

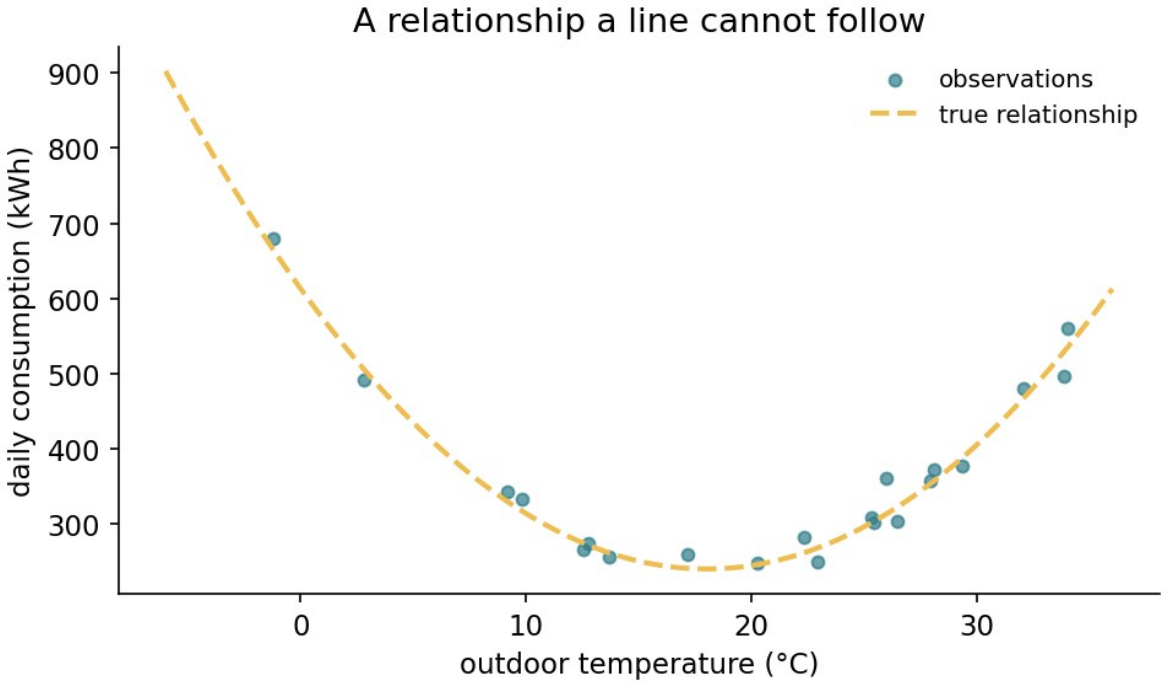
Energy consumption against temperature: heating in the cold, cooling in the heat.

No straight line follows that.

In code: `PolynomialFeatures(degree=2)` adds the squared temperature, $t^2$, as a second column.

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# A relationship a line cannot follow



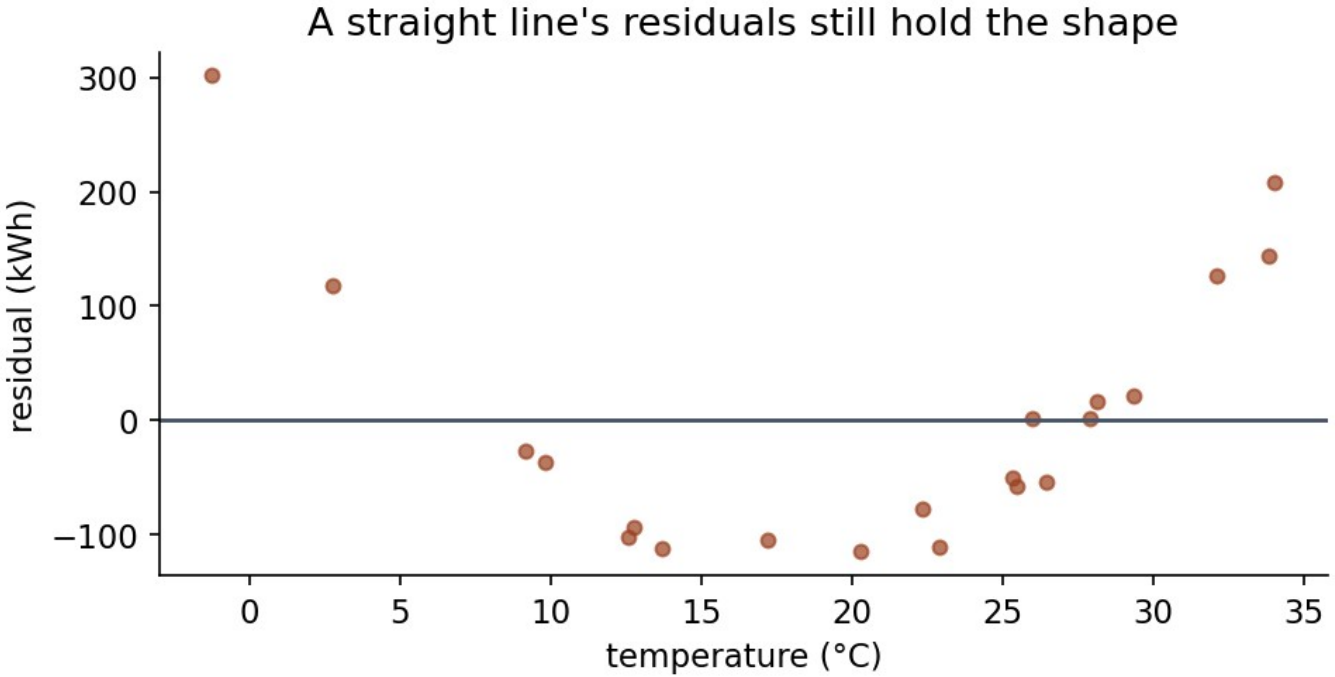

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# “Linear” means linear in the coefficients

$$\hat{y} = w_1 t + w_2 t^2 + b$$

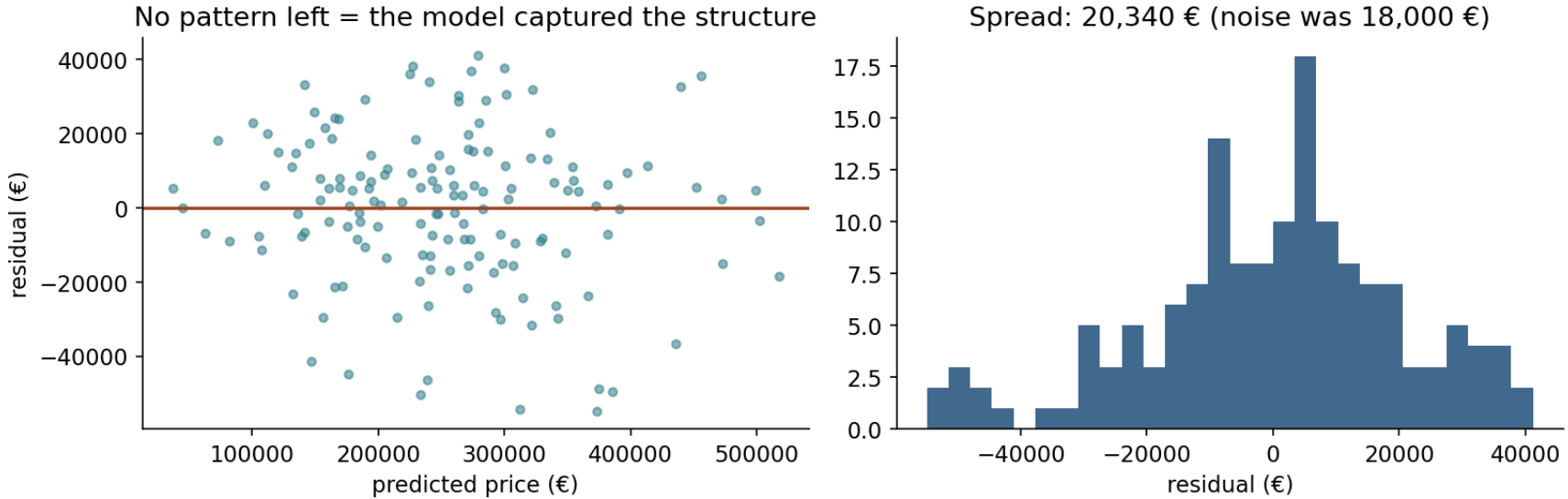
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# The straight line’s residuals keep the shape



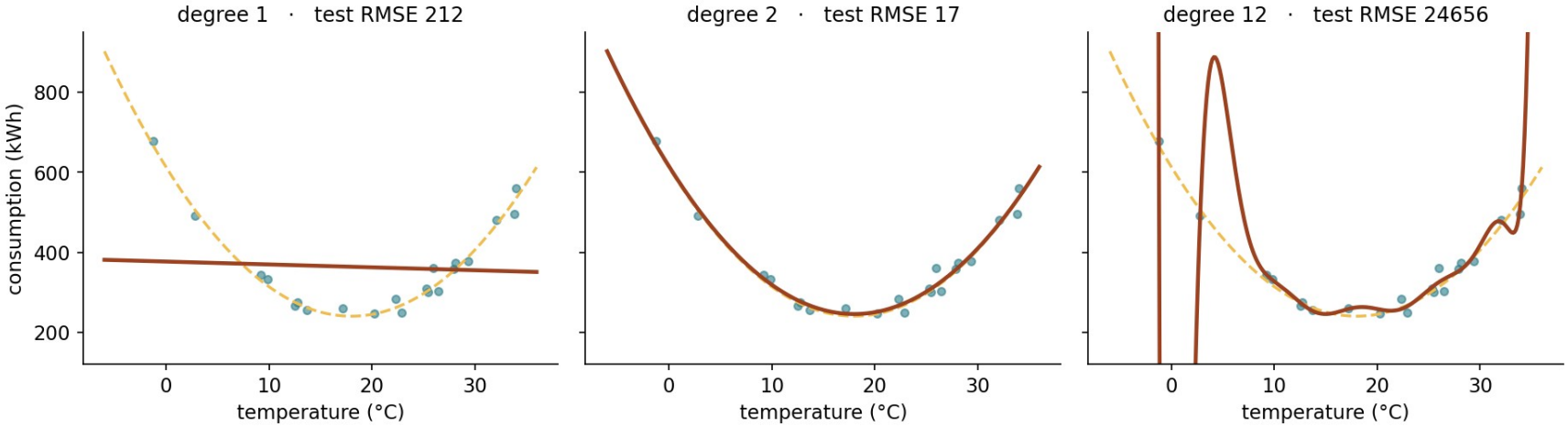

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# What “nothing left” looks like




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# So use more flexibility?



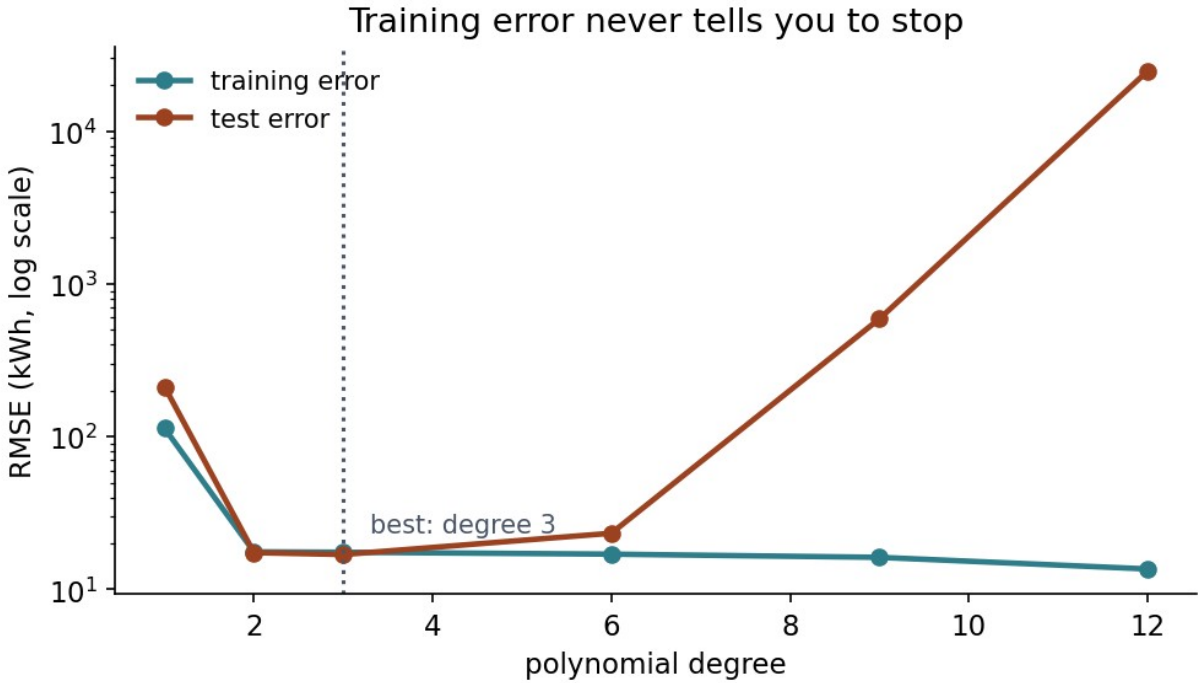

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# Root mean squared error (RMSE), by degree

| Degree | Train RMSE (kWh) | Test RMSE (kWh) |
|---|---|---|
| 1 | 113.3 | 212.1 |
| 3 | 17.5 | **16.9** |
| 9 | 16.2 | 590.1 |
| 12 | 13.6 | 24,655.7 |

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# The gap is the overfitting




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# Notebook 2, live

| section, and its slides | you call (→ the number to come back with) |
|---|---|
| a relationship that bends, 29-30 | `train_test_split` → 21 days to fit, 9 to test |
| linear in the coefficients, 31-33 | `PolynomialFeatures`, `StandardScaler` and `LinearRegression`, chained by `make_pipeline` |
| too much flexibility, 34-36 | `mean_squared_error`, square-rooted → test RMSE 16.9 at degree 3, 24,655.7 at 12 |
| why the wiggles appear, 38 | the fitted coefficients → largest 3,097,038,010 at degree 12 |

# Why flexibility goes wrong

Largest coefficient, degree 2: **411**

Largest coefficient, degree 12: **3,097,038,010**

# Charge for size

$$J_{\text{ridge}} = \text{MSE} + \lambda \sum_j w_j^2 \qquad J_{\text{lasso}} = \text{MSE} + \lambda \sum_j |w_j|$$

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# Two rules that are not optional

- The **intercept is never penalised**: it is not a claim about any feature
- Features **must be scaled first**, or the penalty is arbitrary
- In code: `make_pipeline(StandardScaler(), Ridge(alpha=0.01))`; scikit-learn calls λ `alpha`

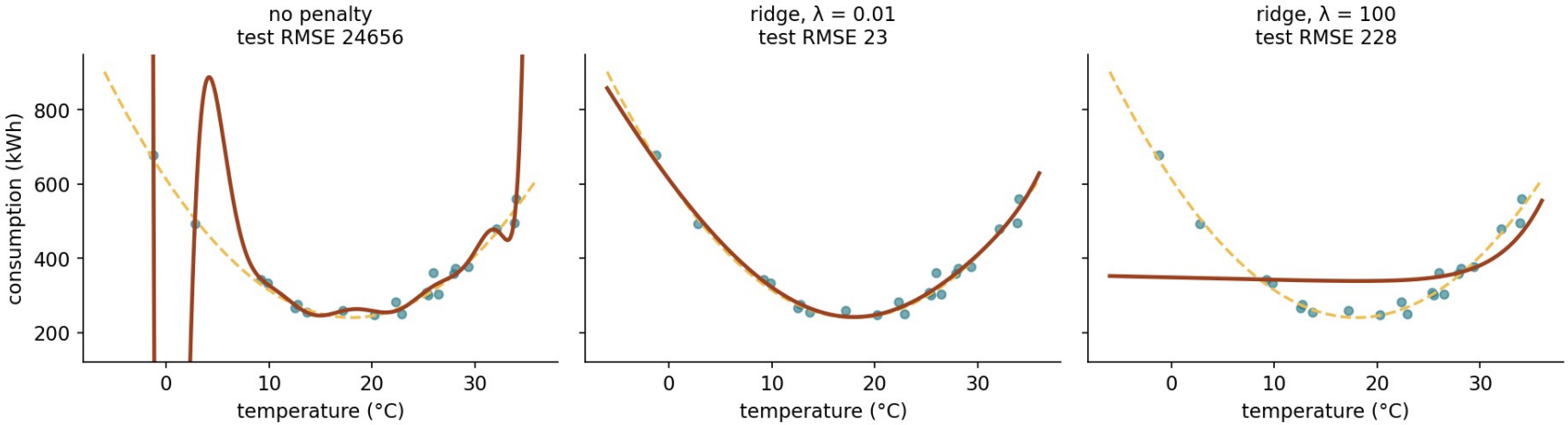
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# What a penalty buys

| Model | Train RMSE | Test RMSE | max \|w\| |
|---|---|---|---|
| none | 13.6 | 24,655.7 | 3,097,038,010 |
| $\lambda = 0.01$ | 18.0 | **22.8** | 365 |
| $\lambda = 1$ | 44.5 | 105.2 | 156 |
| $\lambda = 100$ | 96.0 | 227.6 | 6 |

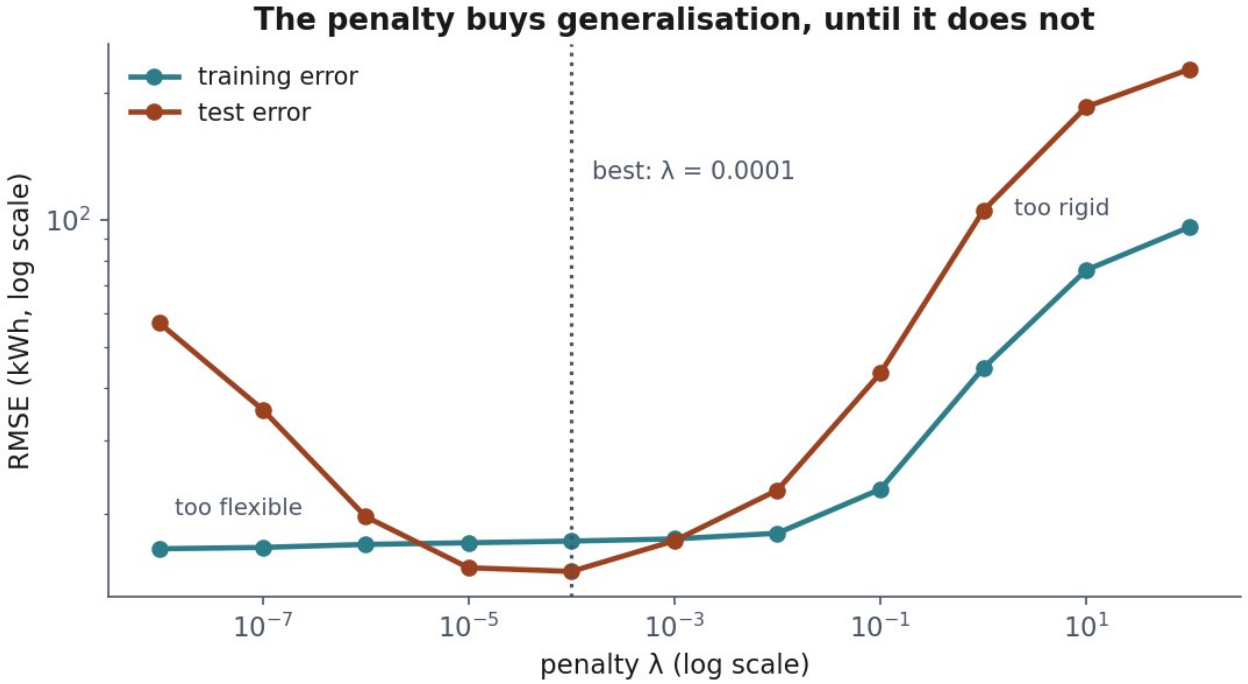
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# The unpenalised fit leaves the picture




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# Too flexible, too rigid




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# Ridge has an exact solution too

$$\theta = (X^\top X + \lambda I)^{-1} X^\top y$$

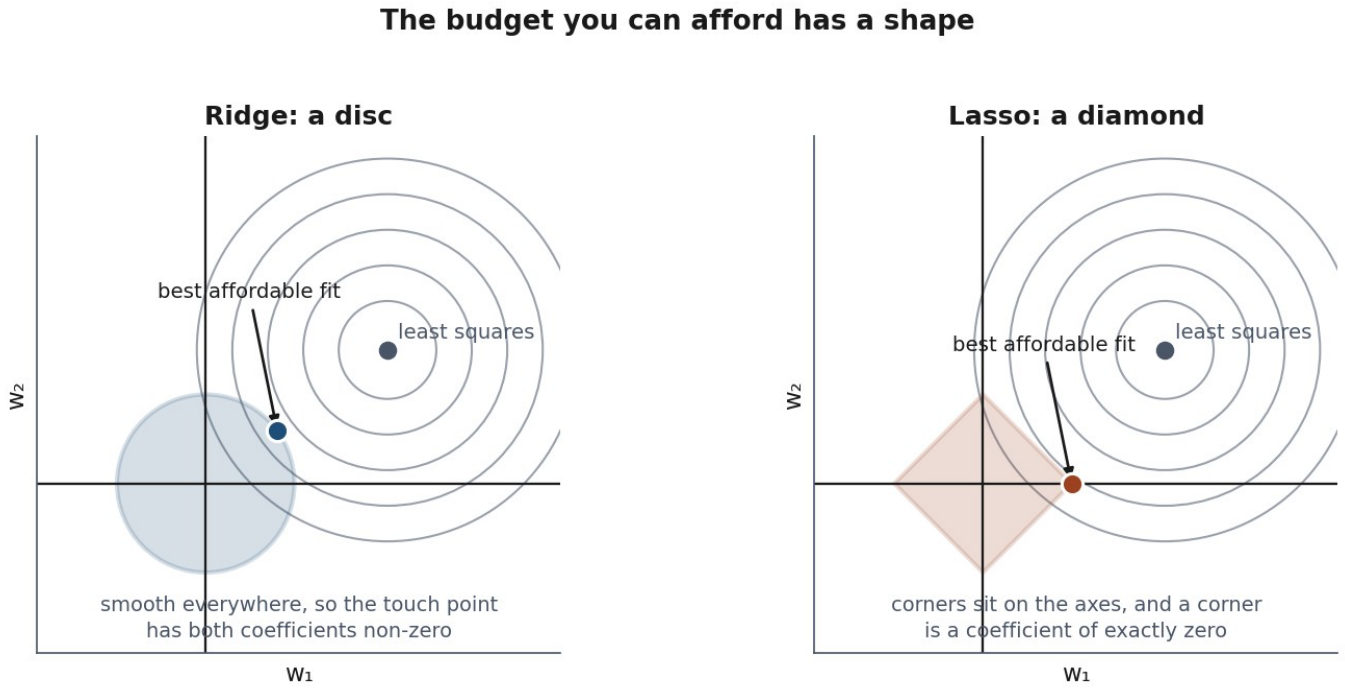
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# Why that one change fixes it

- Least squares fails when a direction costs nothing: the valley is flat
- The penalty makes **every** direction cost something
- Every eigenvalue of $X^TX$ shifts up by $\lambda$, so none of them can be zero

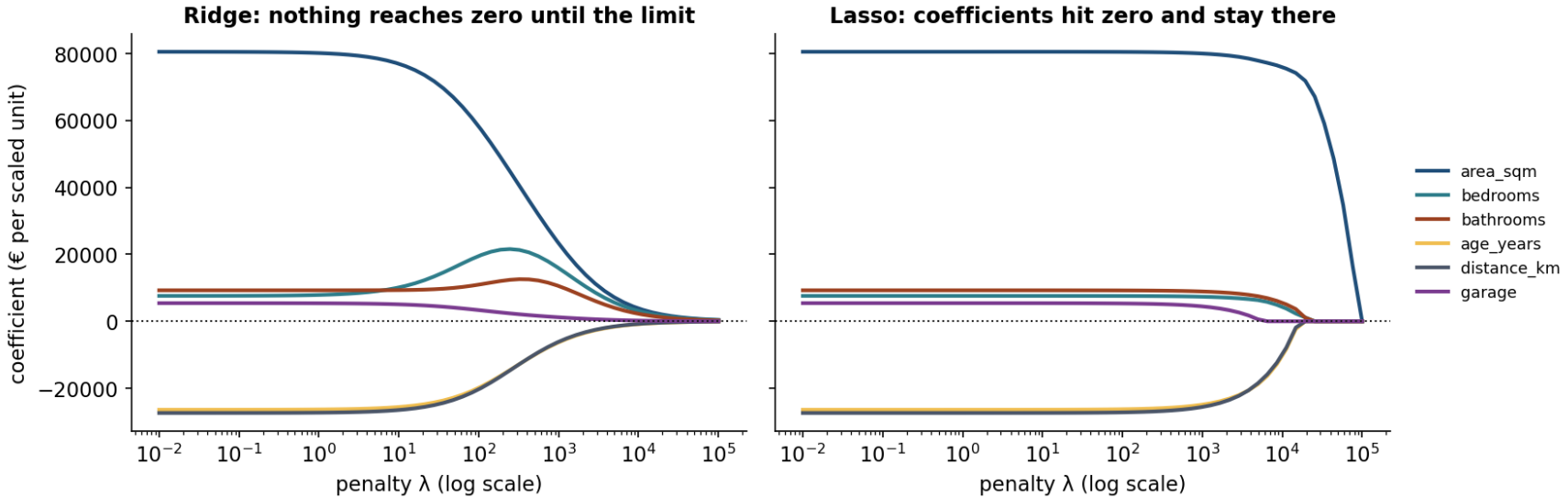
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# The picture behind Ridge vs Lasso




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# Ridge shrinks, Lasso selects




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# The order Lasso drops features

| λ | Features kept |
|---|---|
| 1,000 | all six |
| 10,000 | five: `garage` dropped |
| 20,000 | three: `area_sqm`, `bedrooms`, `bathrooms` |
| 40,000 | one: `area_sqm` |

# The case Ridge was invented for

`area_sqm` and `area_sqft`: the same measurement, two units.

Correlation: **0.999999**

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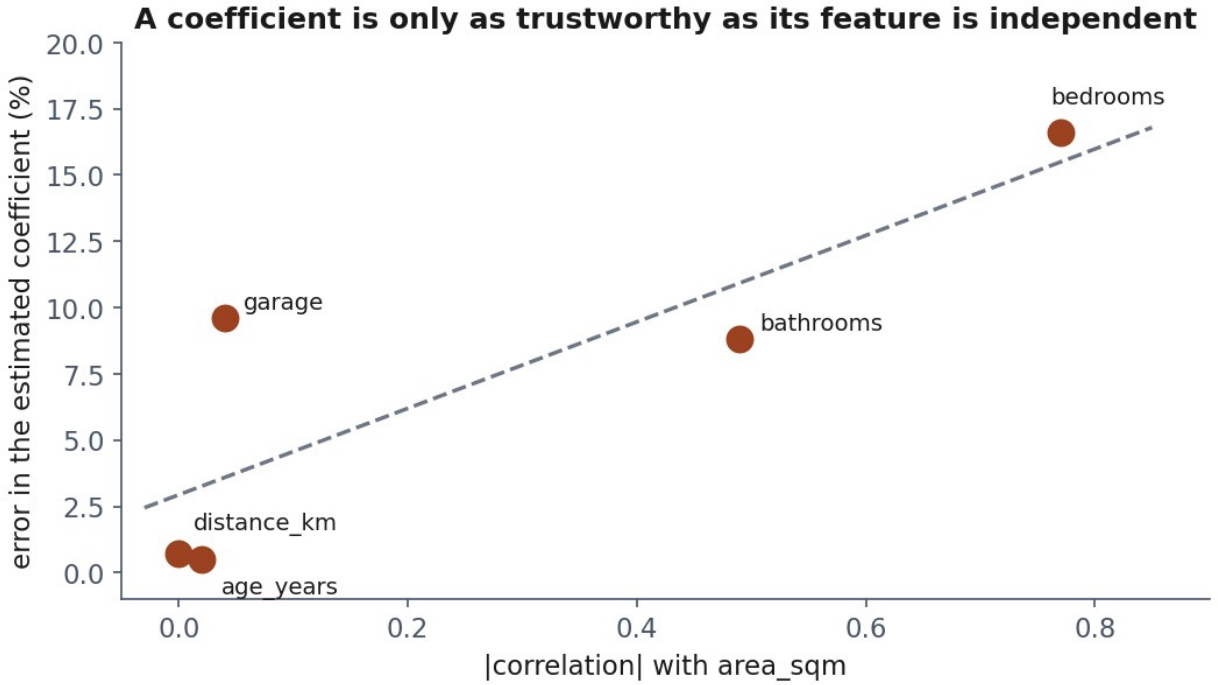
# The coefficients are noise

| Split | area_sqm | area_sqft |
|---|---|---|
| 0 | 9,261 | −640 |
| 1 | 14,535 | −1,127 |
| 2 | 45,307 | −3,989 |
| 3 | 39,061 | −3,402 |

# What a coefficient actually says

The change in the prediction for a one-unit change in that feature, **with every other feature held fixed**.

# When can you trust a coefficient?




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# Notebook 3, live

| section, and its slides | you call (→ the number to come back with) |
|---|---|
| Ridge on the degree-12 fit, 39-43 | `Ridge(alpha=0.01)` after `PolynomialFeatures` and `StandardScaler` → largest coefficient 365 |
| regularisation paths, 46-48 | `make_pipeline(StandardScaler(), Ridge(alpha=...))`, the same with `Lasso` → only area left at 40,000 |
| the collinear case, 49-50 | `LinearRegression` against `Ridge(alpha=10)` on four splits → 9,261 to 45,307, then about 40,000 |

# What we did today

- A model that claims a fixed amount per unit
- An exact solution (a projection), and when it fails
- An iterative one: walk downhill, mind your stride
- Flexibility helps until it does not
- Charging for size buys generalisation, and readable coefficients
- **365** against billions: what a penalty of 0.01 costs, and buys

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# Homework: we discuss it on Friday 23 October

`Exercises/03_regression.md`

Fit, regularise, and **explain which coefficients you believe**.

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